Proseminar

Kalman Filter

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Abstract: Kalman filters and their variants, such as the Extended Kalman Filter(EKF) and Unscented Kalman Filter(UKF), have become one of the most enabling state estimators in dynamic systems, ranging from robotics and aerospace to signal processing. While the Kalman filter itself provides an effective recursive algorithm for linear systems with Gaussian noise, the EKF extends this to non-linear systems through dynamical linearization. The UKF further improves estimation accuracy in non-linear scenarios by utilizing the Unscented Transform, avoiding the need for linearization altogether. This paper shows the historical development and mathematical background of these algorithms, highlighting their key principles, strengths, and limitations.

1 Introduction

In fields like robotics, aerospace, and signal processing, systems often have to work under uncertain and noisy[1] conditions. Whether it's determining the position of a vehicle by using GPS or tracking an aircraft's trajectory, achieving reliable state estimation is essential. Sensors alone cannot provide accurate data due to noise and errors. That is why integrating sensor measurements with predictive models like the Kalman Filter becomes so important.

In 1960, Rudolf E. Kalman proposed the Kalman filter [1], an algorithm designed to solve this problem. The Kalman filter offers a computationally efficient and recursive approach to estimating the state of dynamic systems. By combining predictions from a mathematical model with real-world measurements. Provides an optimal estimate of the current and future state of the system. Its recursive design makes it particularly well-suited for real-time applications, where both speed and accuracy are critical [2] [7].

However, the standard Kalman filter is designed for linear systems[2] with Gaussian noise[3]. While highly effective in these cases, it struggles with non-linear[4] dynamics. To overcome this limitation, the Extended Kalman Filter (EKF) and the Unscented Kalman Filter (UKF) were developed. These methods adapt the principles of the Kalman filter to non-linear systems [5], broadening its use in areas such as robotics, where non-linear behaviors are common.

2 History of Gaussian and Kalman Filters

The development of Gaussian and Kalman filters marks a significant advancement in the field of state estimation. Early methods, like the Bayes filter, laid the groundwork for probabilistic approaches. The Bayes filter provided a way to estimate the state of a system by combining prior knowledge with new measurements. This recursive process continuously refined the estimates as new data became available, effectively handling uncertainty

[1] Refers to uncertainties or errors in measurements and system dynamics.

[2] Systems with outputs directly proportional to inputs, described by linear equations.

[3] Gaussian noise is a type of random noise with a probability distribution that follows a normal (Gaussian) curve, characterized by its mean and variance.

[4] Systems with complex, non-proportional input-output relationships.

in dynamic systems. However, its generality made the practical implementation computationally expensive. The Kalman filter simplified the application of Bayesian principles by assuming Gaussian noise, making real-time computation possible and efficient.

Gaussian filters are based on the assumption that both the system state and noise follow Gaussian distributions. This property simplifies calculations because a Gaussian distribution is fully described by its mean and covariance. These attributes allow for computationally manageable state estimation, even in complex systems. The Gaussian filter became a key tool for addressing noise and uncertainty in practical applications. The work of Dr. Kalman incorporated the properties of the Gaussian distribution in his algorithm, establishing it as the mathematical framework for the Kalman filter.

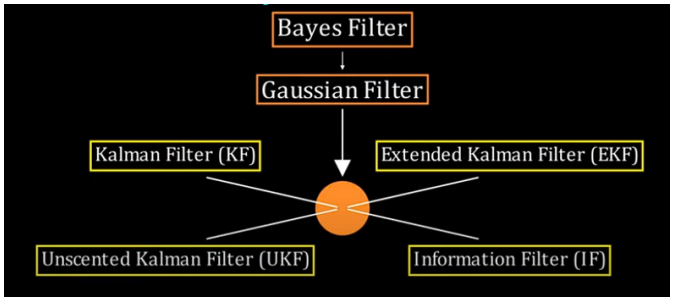


Figure 1: Taxonomy of Gaussian Filter. Image from [8]

The Kalman filter was designed to solve pressing challenges in dynamic systems. Before its invention, state estimation relied on direct measurements or basic methods that were often unreliable in noisy conditions. Dr. Kalman developed a method that could combine model predictions with noisy sensor data to produce optimal state estimates. This innovation was driven by the need for spacecraft navigation, where precise and real-time tracking of position and velocity was critical. The Kalman filter provided the accuracy and speed required for these missions, cementing its role as a vital tool in aerospace and expanding its use to robotics, automotive engineering, and signal processing.

3 Linear Kalman-Filter

3.1 The General Concept of Linear Kalman Filter

The Linear Kalman Filter (LKF) is a recursive algorithm designed to estimate the state of a dynamic system that evolves linearly over time while subject to random noise. Introduced by Rudolf E. Kalman in 1960, this method addresses critical challenges in signal processing, prediction, and noise filtering by providing optimal estimates under linear system assumptions and Gaussian noise [1].

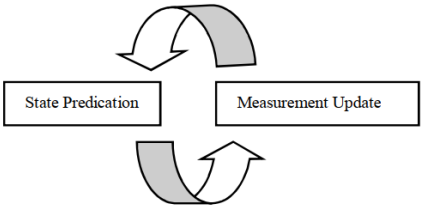


Figure 2: Function of Kalman Filter. Image from [5]

3.2 Prediction and Update Steps

The Kalman Filter operates iteratively through two main stages: Prediction and Update. Each stage refines the system's state estimate while balancing the contributions of the system model and observed measurements.

a) Prediction Step: The prediction step forecasts the system's next state based on its current state and a mathematical model [1][5]. This step has two key parts:

1. **State Prediction:** The system's next state is estimated using the state transition equation:

   $$\hat{x}_{k|k-1} = A\hat{x}_{k-1|k-1} + Bu_{k-1}$$

   Where:

   - $\hat{x}_{k|k-1}$: Predicted state estimate.
   - A: State transition matrix.
   - $\hat{x}_{k-1|k-1}$: Previous state estimate.
   - Bu_{k-1}: External control input.

2. **Covariance Prediction:** The prediction's uncertainty is updated using:

   $$P_{k|k-1} = AP_{k-1|k-1}A^{\top} + Q$$

   Where:

   - $P_{k|k-1}$: Predicted uncertainty (covariance).
   - Q: Process noise covariance.

b) Update Step: The update step improves the predicted state by incorporating new measurements [1][5]. This step involves:

1. **Measurement Residual:** The difference between the observed measurement and the predicted one:

$$y_k = z_k - H\hat{x}_{k|k-1}$$

Where:

- y_k: Measurement Residual.

- z_k: Actual measurement.

- H: Measurement matrix.

2. **Covariance:** The uncertainty of the measurement difference:

$$S_k = HP_{k|k-1}H^\top + R$$

Where:

- R: is the measurement noise covariance.

3. **Optimal Kalman Gain:** The Kalman gain determines how much to trust the measurement versus the prediction:

$$K_k = P_{k|k-1}H^\top S_k^{-1}$$

4. **Update State Estimation:** The predicted state is corrected based on the measurement:

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k y_k$$

5. **Covariance of State Estimation:** The updated uncertainty is given by:

$$P_{k|k} = (I - K_k H)P_{k|k-1}$$

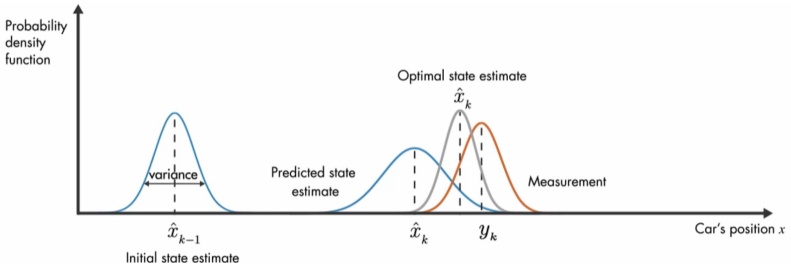


Figure 3: Graphic of Optimal State Estimate from an example of a car's position. Image from [9]

c) Overall Algorithm Flow: As shown in Figure 3, the Kalman filter starts with an initial guess of the system's state $\hat{x}_{k-1}$ and its uncertainty P_{k-1}. At each step, it alternates between the prediction and update stages.

The Kalman filter combines predictions and measurements in a weighted manner using the Kalman gain K_k. This gain decides how much to rely on each source of information. The final estimate is:

$$\hat{x}_{k|k} = K_k y_k + (1 - K_k)\hat{x}_{k|k-1}.$$

This process repeats at every time step, refining the optimal state estimate over time.

3.3 Applications of the Linear Kalman Filter (LKF)

The Linear Kalman Filter (LKF) is widely used in aerospace and navigation, it tracks the position and velocity of vehicles, such as aircraft and spacecraft, using sensor data like GPS to improve accuracy [6]. The LKF is also crucial in signal processing, where it is used for noise reduction, speech enhancement, and communication systems to correct errors in transmitted signals. In finance, it estimates and predicts stock prices and other economic variables by modeling noisy time-series data, helping traders and analysts make more informed decisions. In robotics, the LKF is applied in scenarios with linear motion, such as wheeled robots, for localization, mapping, and path planning, especially when the system dynamics can be approximated linearly.

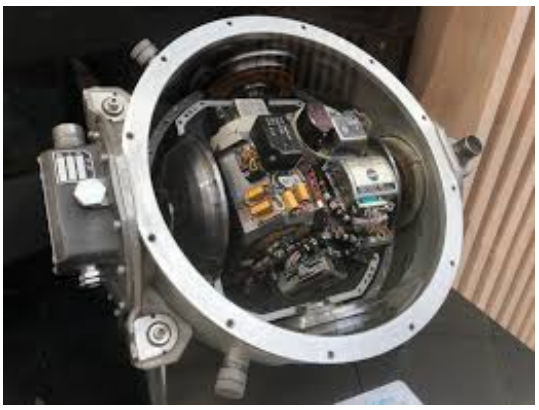


Figure 4: Inertial Measurement Unit (IMU) used in the Apollo missions for Kalman Filter calculations. Image from NASA Images.

3.4 Limitations

Although its widespread use, the Kalman Filter has several limitations. It assumes that the system dynamics and measurement models are linear, which makes it unsuitable for non-linear systems. Additionally, it assumes that both the process and measurement noise

follow Gaussian distributions, which may not always be the case in real-world applications, especially with non-Gaussian noise. The Kalman filter's performance is also sensitive to the accuracy of the system model and noise covariance matrices; poor estimation of these values can lead to inaccurate results. Finally, the Kalman filter requires real-time computation, which can be computationally expensive, particularly in systems with large or complex state spaces.

4 Extended Kalman-Filter (EKF)

4.1 The General Concept of Extended Kalman-Filter

While the Kalman filter works well for linear systems with Gaussian noise, it struggles with non-linear systems. To avoid this, the Extended Kalman Filter (EKF) was developed. The EKF approximates non-linear system behavior by linearizing the dynamics around the current state. This method allows the Kalman filter principles to apply to non-linear systems.

4.2 Linearization Process in the Extended Kalman Filter

The Extended Kalman Filter (EKF) relies on linearization to approximate non-linear system dynamics and measurement models. This step transforms the non-linear equations into a locally linear form using the Jacobian matrices of the system's state transition and measurement functions.

a) System Representation: The EKF operates on non-linear systems described by:

$$x_k = f(x_{k-1}, u_{k-1}) + w_{k-1}$$

$$z_k = h(x_k) + v_k$$

Where:

- $f(x)$ and $h(x)$: Are non-linear functions.

-w_k: Represents process noise.

-v_k: Represents measurement noise.

b) Jacobian Matrices: The non-linear functions $f(x_{k-1}, u_{k-1})$ and $h(x_k)$ are linearized around the current estimate using their first-order Taylor expansion. The resulting **Jacobian matrices** are:

- **State Transition Jacobian:**

$$F_k = \frac{\partial f}{\partial x}\Big|_{x_{k-1}, u_{k-1}}$$

This matrix captures how the state evolves with small changes in x_{k-1}.

- **Measurement Jacobian:**

$$H_k = \frac{\partial h}{\partial x}\Big|_{x_k}$$

This matrix describes the relationship between the state x_k and the measurements z_k.

c) Linearized Prediction and Update: Using these Jacobian matrices, the EKF performs its recursive steps:

- **Prediction:**

$$\hat{x}_{k|k-1} = f(\hat{x}_{k-1|k-1}, u_{k-1})$$

$$P_{k|k-1} = F_k P_{k-1|k-1} F_k^\top + Q_k$$

Where:

- $\hat{x}_{k|k-1}$: The predicted state.

- $P_{k|k-1}$: The predicted covariance.

- **Update:**

$$y_k = z_k - h(\hat{x}_{k|k-1})$$

$$S_k = H_k P_{k|k-1} H_k^\top + R_k$$

$$K_k = P_{k|k-1} H_k^\top S_k^{-1}$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k y_k$$

$$P_{k|k} = (I - K_k H_k) P_{k|k-1}$$

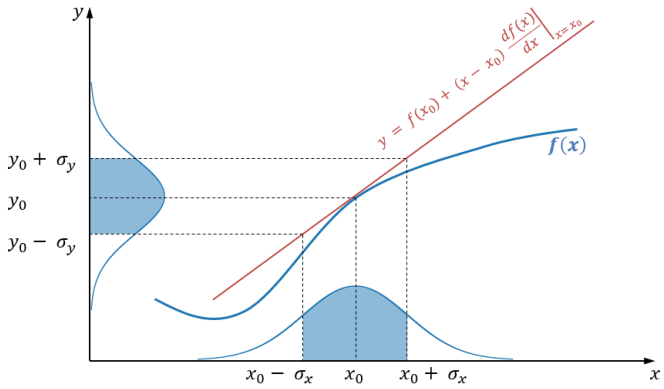


Figure 5: Image of Linearization."f(x)" non-linear function of the system, "y" linearized function of the system. Image from Kalman Filter, Alex Becker

d) Purpose of Linearization: The linearization process allows the EKF to apply Kalman filter principles to non-linear systems by approximating them locally as linear. While effective for systems with mild non-linearities, the approximation can introduce inaccuracies when non-linearities are significant [5].

4.3 Limitations of the Extended Kalman Filter

The Extended Kalman Filter (EKF) has notable limitations that can impact its performance. It relies on a first-order Taylor series expansion to linearize non-linear functions, which introduces errors in highly non-linear systems. The need to compute Jacobian matrices (F_k and H_k) at every iteration increases computational complexity and can be error-prone, especially for complex systems. Additionally, the EKF assumes Gaussian noise for both the process and measurements, making it less effective when noise deviates from this assumption. Its performance is highly sensitive to the initial state estimate ($\hat{x}_{k-1}$) and covariance (P_{k-1}); poor initialization can lead to divergence. The local linearization process assumes the state estimate is close to the true state, which may not hold in cases of poor initialization or strongly non-linear dynamics. These limitations, coupled with high computational demands for matrix operations, have led to the development of alternatives like the Unscented Kalman Filter (UKF) that address many of these challenges [5].

5 Unscented Kalman-Filter (UKF)

5.1 General Concept of the Unscented Kalman Filter (UKF)

The Unscented Kalman Filter (UKF) is a recursive algorithm designed for state estimation in non-linear systems. Unlike the Extended Kalman Filter (EKF), the UKF avoids the need for linearization of system and measurement models. Instead, it employs the Unscented Transform (UT) to propagate mean and covariance through non-linear transformations, capturing higher-order effects and improving accuracy in non-linear systems[5][7].

5.2 Key Principles of the UKF

1. **Sigma Point Generation:** The UKF approximates the state distribution using $2n+1$ sigma points for an n-dimensional state. These points are chosen deterministically around the state mean and capture the mean and covariance of the state distribution. The sigma points are computed as:

   $$\chi_0 = \hat{x}, \quad \chi_i = \hat{x} + \sqrt{(n+\lambda)P}_i, \quad \chi_{i+n} = \hat{x} - \sqrt{(n+\lambda)P}_i$$

   Where:

   - $\lambda = \alpha^2(n+\kappa) - n$: Scaling parameter.
   - α: Controls the spread of the sigma points.
   - κ: Cecondary scaling parameter.
   - P: The covariance matrix.

2. **Non-linear Propagation:** Each sigma point is passed through the non-linear state transition function:

   $$\chi_i' = f(\chi_i, u)$$

   where $f(\cdot)$ represents the non-linear dynamics of the system. This step transforms the sigma points while preserving their statistical properties.

3. **Predicted State and Covariance:** The transformed sigma points are used to compute the predicted state mean and covariance:

   $$\hat{x}_{k|k-1} = \sum_{i=0}^{2n} W_i \chi_i'$$

   $$P_{k|k-1} = \sum_{i=0}^{2n} W_i \left(\chi_i' - \hat{x}_{k|k-1}\right) \left(\chi_i' - \hat{x}_{k|k-1}\right)^\top + Q$$

   where W_i are the weights for the sigma points, and Q is the process noise covariance.

4. **Measurement Update Using Sigma Points:** The sigma points are propagated through the non-linear measurement function h(x):

   $$\gamma_i = h(\chi_i')$$

   The predicted measurement mean and covariance is computed as:

   $$z_k = \sum_{i=0}^{2n} W_i \gamma_i$$

   $$S_k = \sum_{i=0}^{2n} W_i (\gamma_i - z_k)(\gamma_i - z_k)^\top + R$$

   Where:

   - R: The measurement noise covariance.

5. **State Update:** The cross-covariance between the state and measurement is computed as:

   $$P_{xz} = \sum_{i=0}^{2n} W_i (\chi_i' - \hat{x}_{k|k-1})(\gamma_i - z_k)^\top$$

   Using the Kalman gain, the updated state and covariance are computed:

   $$K_k = P_{xz} S_k^{-1}$$

   $$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k (z_k - z_k^{pred})$$

   $$P_{k|k} = P_{k|k-1} - K_k S_k K_k^\top$$

The UKF addresses the shortcomings of linearization by propagating uncertainty directly through non-linear functions. Its use of sigma points ensures higher accuracy and robustness, making it a reliable alternative to the EKF for non-linear systems [7].

5.3 Limitations of the Unscented Kalman Filter (UKF)

While the Unscented Kalman Filter (UKF) improves accuracy over the Extended Kalman Filter, it has still limitations. The computational cost is higher due to the generation and propagation of sigma points ($2n + 1$) and matrix operations, making it less suitable for high-dimensional systems. The assumption of Gaussian noise and state distributions limits its accuracy in non-Gaussian scenarios. Performance is sensitive to parameters like α, β, and κ, which require careful tuning. Additionally, numerical stability issues can arise during matrix square root computations, particularly with poorly conditioned covariance matrices [7].

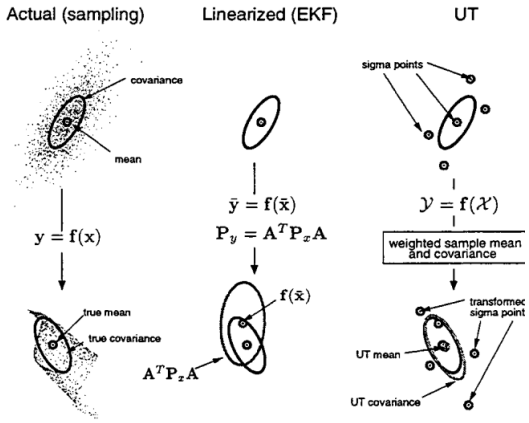


Figure 6: Example of the Unscented Transformation (UT) for mean and covariance propagation. a) actual, b) first-order linearization (EKF), c) UT. Image from [7].

6 Conclusions

The Kalman filter is plays a crucial role in state estimation for dynamic systems[1]. The original Kalman filter works well for linear systems with Gaussian noise, but as systems became more complex, the EKF and UKF were developed to handle non-linear systems. The EKF linearizes non-linearities, but the UKF avoids linearization and provides better accuracy in highly non-linear systems[7].

Despite their strengths, both the EKF and UKF face limitations, such as the need for careful parameter tuning, computational complexity, and the assumption of Gaussian noise. These filters may struggle in real-world scenarios with non-Gaussian noise or highly uncertain dynamics [5].

The future of Kalman filtering lies in reducing computational costs, enhancing robustness, and combining traditional methods with new approaches, which will expand their use in more complex systems [2].

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